

A novel feature selection method for twin support vector machine



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ABSTRACT

Both support vector machine (SVM) and twin support vector machine (TWSVM) are powerful classification tools. However, in contrast to many SVM-based feature selection methods, TWSVM has not any corresponding one due to its different mechanism up to now. In this paper, we propose a feature selection method based on TWSVM, called FTSVM. It is interesting because of the advantages of TWSVM in many cases. Our FTSVM is quite different from the SVM-based feature selection methods. In fact, linear SVM constructs a single separating hyperplane which corresponds a single weight for each feature, whereas linear TWSVM constructs two fitting hyperplanes which corresponds to two weights for each feature. In our linear FTSVM, in order to link these two fitting hyperplanes, a feature selection matrix is introduced. Thus, the feature selection becomes to find an optimal matrix, leading to solve a multi-objective mixed-integer programming problem by a greedy algorithm. In addition, the linear FTSVM has been extended to the nonlinear case. Furthermore, a feature ranking strategy based on FTSVM is also suggested. The experimental results on several public available benchmark datasets indicate that our FTSVM not only gives nice feature selection on both linear and nonlinear cases but also improves the performance of TWSVM efficiently.

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1. Introduction

Feature selection [18,15,6,11,19] is an important problem since it removes the irrelevant features and maintains the relevant features that are as close to the class as possible. The benefit of feature selection is twofold. On one hand, it is meaningful because it can identify the features that contribute most to classification. On the other hand, feature selection is helpful for solving the classification problem since it can not only reduce the dimension of input space and speed up the computation procedure but also improve the classification accuracy. There are mainly two types of feature selection methods: one is the general methods independent of any classifiers, e.g., Fisher score [38] and Laplacian score [12]; the other one is the wrapper-type method dependent on the classifier, e.g., the methods based on Bayesian network [14], based on neural networks [32] and based on support vector machine (SVM) [2,23]. The wrapper-type methods attract more attentions and often improve the performance of the original classifier as well [14,37,23,2,32].

It is interesting to investigate the wrapper-type feature selection methods based on the twin support vector machine (TWSVM) [16,35]. TWSVM generates two nonparallel fitting hyperplanes and

has superior performance than SVM [3,21,7,43] on both the classification accuracy and learning speed in many practical applications [16,35,33,30]. Particularly, it is suitable for some special datasets, e.g., “cross-planes” [16]. However, in contrast to SVM which owns many wrapper-type feature selection methods such as SVM-RFE [23] and RFSVM [2], TWSVM has no any corresponding one. The reason is that the current SVM-based feature selection methods cannot be used to TWSVM directly. In fact, for each feature, SVM provides a single weight corresponding to the single separating hyperplane whereas TWSVM provides two weights corresponding to two fitting hyperplanes, leading to some difficulties for feature selection.

In this paper, we propose a novel feature selection method for TWSVM, called FTSVM for short. Our FTSVM includes two forms: linear FTSVM and nonlinear one. The former is formulated by the following steps: first of all, a basic L_1 -TWSVM is proposed by introducing the L_1 -norm regularization terms, due to the success in L_1 -SVM to obtain a sparse feature weight [44,20,8,13]; then, a feature selection matrix, a diagonal matrix with element either 1 or 0, is introduced in the proposed L_1 -TWSVM, resulting to two mixed integer programming problems (MIPPs); finally, the two MIPPs are solved simultaneously as a multi-objective mixed integer programming problem (MOMIPP) [39] by an greedy algorithm. The nonlinear FTSVM is constructed by kernel trick [16,35], which

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needs also to solve a MOMIPP by the proposed greedy algorithm. In addition, based on FTSVM, a feature ranking strategy is suggested, which ranks the features according to their contributions to the objective in the MOMIPP. The experimental results show that our FTSVM not only gives nice feature selection but also improves the performance of TWSVM.

This paper is organized as follows. A briefly review of L_2 -TWSVM is in Section 2, and the standard L_1 -TWSVM is proposed in Section 3. Our feature selection and ranking methods are formulated in Section 4, and experiments are arranged in Section 5. Finally, Section 6 gives the conclusions. For convenience, in Table 1, we present some notations used in the paper.

2. Review of L_2 -TWSVM

Consider the binary classification problem with m_1 training samples belong to positive class represented by A_1 and m_2 training samples belong to negative class represented by A_2 in the n -dimensional real space R^n , a classifier attempts to predict the new samples belong to either positive or negative class.

TWSVM [16] seeks two nonparallel hyperplanes

$$f_1(x) = xw_1 + b_1 = 0 \quad \text{and} \quad f_2(x) = xw_2 + b_2 = 0 \quad (1)$$

such that each one is the fitting hyperplane of one of the positive and negative classes, a new sample will be predicted to one class if it is closer to the corresponding fitting hyperplane. TWSVM keeps the fitting hyperplane as close as possible to its corresponding training samples and far away from the others. The purpose of L_2 -TWSVM [35] is very the same as TWSVM, where the difference is L_2 -TWSVM introduces the L_2 regularization terms of the weight vectors into TWSVM to minimize the structural risk. L_2 -TWSVM solves following quadratic programming problems (QPPs)

$$\begin{aligned} \min_{w_1, b_1, \xi_1} \quad & \frac{1}{2} (\|w_1\|_2^2 + b_1^2 + c_{11} \|A_1 w_1 + b_1 e\|_2^2) + c_{12} e^T \xi_1 \\ \text{s.t.} \quad & -(A_2 w_1 + b_1 e) + \xi_1 \geq e, \quad \xi_1 \geq 0, \end{aligned} \quad (2)$$

and

$$\begin{aligned} \min_{w_2, b_2, \xi_2} \quad & \frac{1}{2} (\|w_2\|_2^2 + b_2^2 + c_{21} \|A_2 w_2 + b_2 e\|_2^2) + c_{22} e^T \xi_2 \\ \text{s.t.} \quad & A_1 w_2 + b_2 e + \xi_2 \geq e, \quad \xi_2 \geq 0, \end{aligned} \quad (3)$$

where $c_{11}, c_{12}, c_{21}, c_{22} > 0$ are parameters, $\xi_1 \in R^{m_1}$ and $\xi_2 \in R^{m_2}$ are slack vectors.

Table 1
The symbols used in this paper.

Symbol	Domain	Description
m_1, m_2	Z^+	Number of positive and negative training samples
m	Z^+	Number of training samples, $m = m_1 + m_2$
n	Z^+	Dimension of training samples
x_i	R^n	The i th training sample
y_i	$\{+1, -1\}$	Class of the i th training sample
A_1, A_2	$R^{m_1 \times n}, R^{m_2 \times n}$	Positive and negative training matrices
A	$R^{m \times n}$	Training matrix, $A = [A_1; A_2]$
e		Vector of ones of appropriate dimension
$w_{1,2}, b_{1,2}$	R^n, R	Normal vectors and biases
$u_{1,2}, \gamma_{1,2}$	R^m, R	Normal vectors and biases
$\xi_{1,2}$	$R^{m_{1,2}}$	Slack vectors
$c_{11,12,21,22}$	R^+	Parameters
σ	R^+	Parameter
λ	$(0, 1)$	Parameter
E	$\text{diag}(1 \text{ or } 0)$	Diagonal matrix
$K(\cdot, \cdot)$		Kernel function
$\ \cdot\ _{1,2}$		$L_{1,2}$ norm
$(\cdot)_+$		Replaces the negative elements with 0

The geometric meanings of (2) and (3) are clear. For example, for (2), its objective function makes the fitting hyperplane $f_1(x) = 0$ of the positive class fit the positive class samples A_1 , while the constraints keep the negative class far from this hyperplane to some extent (the bias of each negative training sample to $f_1(x) = 0$ is no more than -1).

The above linear L_2 -TWSVM has been extended to nonlinear classifier by kernel trick [35]. Define the inner product by the kernel function $K(\cdot, \cdot)$, nonlinear L_2 -TWSVM seeks two kernel generated surfaces

$$f_1(x) = K(x, A)u_1 + \gamma_1 = 0 \quad \text{and} \quad f_2(x) = K(x, A)u_2 + \gamma_2 = 0, \quad (4)$$

where A is the whole training set, including A_1 and A_2 .

The corresponding QPPs solved in nonlinear L_2 -TWSVM are

$$\begin{aligned} \min_{u_1, \gamma_1, \xi_1} \quad & \frac{1}{2} (\|u_1\|_2^2 + \gamma_1^2 + c_{11} \|K(A_1, A)u_1 + \gamma_1 e\|_2^2) + c_{12} e^T \xi_1 \\ \text{s.t.} \quad & -(K(A_2, A)u_1 + \gamma_1 e) + \xi_1 \geq e, \quad \xi_1 \geq 0, \end{aligned} \quad (5)$$

and

$$\begin{aligned} \min_{u_2, \gamma_2, \xi_2} \quad & \frac{1}{2} (\|u_2\|_2^2 + \gamma_2^2 + c_{21} \|K(A_2, A)u_2 + \gamma_2 e\|_2^2) + c_{22} e^T \xi_2 \\ \text{s.t.} \quad & K(A_1, A)u_2 + \gamma_2 e + \xi_2 \geq e, \quad \xi_2 \geq 0. \end{aligned} \quad (6)$$

3. L_1 -TWSVM

As stated above, L_1 -SVM which is sparser than L_2 -SVM [8,13] is proposed by replacing the L_2 norm regularization term in L_2 -SVM with L_1 norm. Therefore, we propose the L_1 -TWSVM that minimizes L_1 -norm of the feature weight vectors so that the features are sparser than L_2 -TWSVM. Similar to L_1 -SVM, we introduce the L_1 -norm of $w_i, i = 1, 2$, and straightly transform the other parts in L_2 -TWSVM from L_2 norm to L_1 -norm as follows

$$\begin{aligned} \min_{w_1, b_1, \xi_1} \quad & \|w_1\|_1 + c_{11} \|A_1 w_1 + b_1 e\|_1 + c_{12} \|\xi_1\|_1 \\ \text{s.t.} \quad & -(A_2 w_1 + b_1 e) + \xi_1 \geq e, \quad \xi_1 \geq 0, \end{aligned} \quad (7)$$

and

$$\begin{aligned} \min_{w_2, b_2, \xi_2} \quad & \|w_2\|_1 + c_{21} \|A_2 w_2 + b_2 e\|_1 + c_{22} \|\xi_2\|_1 \\ \text{s.t.} \quad & A_1 w_2 + b_2 e + \xi_2 \geq e, \quad \xi_2 \geq 0. \end{aligned} \quad (8)$$

Different from solving two dual QPPs in L_2 -TWSVM [35], the above problems can be solved as two differentiable linear programming problems (DLPPs) [9,22]. Define $w_i = p_i - q_i$, $A_i w_i + b_i e = s_i - t_i$, $i = 1, 2$, then (7) and (8) are equivalent to

$$\begin{aligned} \min_{p_1, q_1, s_1, t_1, b_1, \xi_1} \quad & e^T (p_1 + q_1) + c_{11} e^T (s_1 + t_1) + c_{12} e^T \xi_1 \\ \text{s.t.} \quad & A_1 (p_1 - q_1) + b_1 e = s_1 - t_1, \\ & A_2 (p_1 - q_1) + b_1 e \leq -e + \xi_1, \\ & p_1, q_1, s_1, t_1, \xi_1 \geq 0, \end{aligned} \quad (9)$$

and

$$\begin{aligned} \min_{p_2, q_2, s_2, t_2, b_2, \xi_2} \quad & e^T (p_2 + q_2) + c_{21} e^T (s_2 + t_2) + c_{22} e^T \xi_2 \\ \text{s.t.} \quad & A_2 (p_2 - q_2) + b_2 e = s_2 - t_2, \\ & A_1 (p_2 - q_2) + b_2 e \geq e - \xi_2, \\ & p_2, q_2, s_2, t_2, \xi_2 \geq 0. \end{aligned} \quad (10)$$

Note $s_i = A_i(p_i - q_i) + b_i e + t_i, i = 1, 2$, the final problems we solved are

$$\begin{aligned}
\min_{p_1, q_1, t_1, b_1, \xi_1} \quad & e^\top(p_1 + q_1) + c_{11}e^\top(A_1(p_1 - q_1) + 2t_1) + c_{11}m_1b_1 + c_{12}e^\top\xi_1 \\
\text{s.t.} \quad & -A_1(p_1 - q_1) - b_1e - t_1 \leq 0, \\
& A_2(p_1 - q_1) + b_1e - \xi_1 \leq -e, \\
& p_1, q_1, t_1, \xi_1 \geq 0,
\end{aligned} \tag{11}$$

and

$$\begin{aligned}
\min_{p_2, q_2, t_2, b_2, \xi_2} \quad & e^\top(p_2 + q_2) + c_{21}e^\top(A_2(p_2 - q_2) + 2t_2) + c_{21}m_2b_2 + c_{22}e^\top\xi_2 \\
\text{s.t.} \quad & -A_2(p_2 - q_2) - b_2e - t_2 \leq 0, \\
& -A_1(p_2 - q_2) - b_2e - \xi_2 \leq -e, \\
& p_2, q_2, t_2, \xi_2 \geq 0.
\end{aligned} \tag{12}$$

The DLPPs (11) and (12) can be solved by many algorithms and toolboxes. In our experiments, we used a variation of the well known simplex algorithm for linear programming problem [5].

For nonlinear case, our L_1 -norm nonlinear TWSVM also seeks two kernel generated surfaces the same as (4). The corresponding problems are as follows:

$$\begin{aligned}
\min_{u_1, \gamma_1, \xi_1} \quad & \|u_1\|_1 + c_{11}\|K(A_1, A)u_1 + \gamma_1e\|_1 + c_{12}\|\xi_1\|_1 \\
\text{s.t.} \quad & -(K(A_2, A)u_1 + \gamma_1e) + \xi_1 \geq e, \quad \xi_1 \geq 0,
\end{aligned} \tag{13}$$

and

$$\begin{aligned}
\min_{u_2, \gamma_2, \xi_2} \quad & \|u_2\|_1 + c_{21}\|K(A_2, A)u_2 + \gamma_2e\|_1 + c_{22}\|\xi_2\|_1 \\
\text{s.t.} \quad & K(A_1, A)u_2 + \gamma_2e + \xi_2 \geq e, \quad \xi_2 \geq 0,
\end{aligned} \tag{14}$$

where $K(\cdot, \cdot)$ is the kernel function the same as nonlinear L_2 -TWSVM. The problems (13) and (14) could be converted to two DLPPs the same as the linear case if $K(A_i, A)$ in (13) or (14) is replaced with A_i in (7) or (8), $i = 1, 2$, respectively.

L_1 -TWSVM has at least two advantages: (1) the sparsity of the solution and the simplicity of learning process, but without loss the generalization of L_2 -TWSVM; (2) L_1 -norm linear TWSVM can be directly as a feature selection method (i.e., if two weights of a feature in the weight vectors w_1 and w_2 are equal to zeros, then the corresponding feature should be deleted). However, if we use the above L_1 -TWSVM to do feature selection directly, the efficiency may not be expected as L_1 -SVM, this is because if one of the two weights of a feature is not zero, then this feature should not be deleted even if the other weight is zero. In addition, the procedure of feature selection of linear L_1 -TWSVM is not suitable for the nonlinear case because of the unknown feature weight. These drawbacks restrict the ability of feature selection of L_1 -TWSVM.

4. Feature selection and ranking

In this section, we shall elaborate a feature selection method, named FTSVM, and a feature ranking method, named RTSVM, based on L_1 -TWSVM.

4.1. FTSVM

In order to select features based on L_1 -TWSVM, we introduce a feature selection matrix $E = \text{diag}(1 \text{ or } 0)$ and seek two nonparallel hyperplanes

$$g_1(x) = (xE)w_1 + b_1 = 0 \quad \text{and} \quad g_2(x) = (xE)w_2 + b_2 = 0. \tag{15}$$

The problems (7) and (8) become to following two mixed integer programming problems (MIPPs)

$$\begin{aligned}
\min_{w_1, b_1, \xi_1, E} \quad & \|w_1\|_1 + c_{11}\|A_1Ew_1 + b_1e\|_1 + c_{12}e^\top\xi_1 + \sigma e^\top Ee \\
\text{s.t.} \quad & -(A_2Ew_1 + b_1e) \geq e - \xi_1, \quad \xi_1 \geq 0, \\
& E = \text{diag}(1 \text{ or } 0),
\end{aligned} \tag{16}$$

and

$$\begin{aligned}
\min_{w_2, b_2, \xi_2, E} \quad & \|w_2\|_1 + c_{21}\|A_2Ew_2 + b_2e\|_1 + c_{22}e^\top\xi_2 + \sigma e^\top Ee \\
\text{s.t.} \quad & A_1Ew_2 + b_2e \geq e - \xi_2, \quad \xi_2 \geq 0, \\
& E = \text{diag}(1 \text{ or } 0),
\end{aligned} \tag{17}$$

where $\sigma \geq 0$ is a parameter to control the number of selected features (since $e^\top Ee = \text{trace}(E)$).

The geometric meanings of problems (16) and (17) are clear. In fact, the feature selection matrix E defines a subspace spanned by the selected features. Therefore, the training samples A_1 and A_2 distribute in different subspace denoted for different E . The objective in (16) makes the hyperplanes $g_1(x) = 0$ closer to the positive class A_1 in this subspace, while the constraints make $g_1(x) = 0$ far from the negative class A_2 to some extent in this subspace, as does in (17).

Note that the above two problems (16) and (17) share the same matrix E but have different objectives, so they should be considered as a multi-objective MIPP (MOMIPP) [39].

$$\begin{cases} \min_{w_1, b_1, \xi_1, E} & \|w_1\|_1 + c_{11}\|A_1Ew_1 + b_1e\|_1 + c_{12}e^\top\xi_1 + \sigma e^\top Ee \\ \min_{w_2, b_2, \xi_2, E} & \|w_2\|_1 + c_{21}\|A_2Ew_2 + b_2e\|_1 + c_{22}e^\top\xi_2 + \sigma e^\top Ee \\ \text{s.t.} & -(A_2Ew_1 + b_1e) \geq e - \xi_1, \quad \xi_1 \geq 0, \\ & A_1Ew_2 + b_2e \geq e - \xi_2, \quad \xi_2 \geq 0, \\ & E = \text{diag}(1 \text{ or } 0), \end{cases} \tag{18}$$

Once the solution of (18) is obtained, one can easily get the selected features from the feature selection matrix E . In addition, a new sample can be predicted by (15).

Before solving the above problem (18), let us extend it to nonlinear case by kernel trick. Corresponding to the kernel generated surfaces (4) in nonlinear L_1 -TWSVM, our nonlinear FTSVM seeks two kernel generated surfaces with a feature selection matrix E as follows

$$\begin{aligned}
h_1(x) &= K(xE, AE)u_1 + \gamma_1 = 0 \quad \text{and} \quad h_2(x) \\
&= K(xE, AE)u_2 + \gamma_2 = 0.
\end{aligned} \tag{19}$$

Similarly, this also leads a MOMIPP

$$\begin{cases} \min_{u_1, \gamma_1, \eta_1, E} & \|u_1\|_1 + c_{11}\|K(A_1E, AE)u_1 + \gamma_1e\|_1 + c_{12}\|\eta_1\|_1 + \sigma e^\top Ee \\ \min_{u_2, \gamma_2, \eta_2, E} & \|u_2\|_1 + c_{21}\|K(A_2E, AE)u_2 + \gamma_2e\|_1 + c_{22}\|\eta_2\|_1 + \sigma e^\top Ee \\ \text{s.t.} & -(K(A_2E, AE)u_1 + \gamma_1e) \geq e - \eta_1, \quad \eta_1 \geq 0, \\ & K(A_1E, AE)u_2 + \gamma_2e \geq e - \eta_2, \quad \eta_2 \geq 0, \\ & E = \text{diag}(1 \text{ or } 0). \end{cases} \tag{20}$$

4.2. Approach to solve the MOMIPP

Without loss of generality, we just consider to solve the nonlinear one (20), and the linear one can be obtained as a special case by using a linear kernel. As remarked earlier, the prediction of a sample by FTSVM is on the basis of the proximity between the sample and two surfaces $h_{1,2}(x) = 0$, and each objective in the MOMIPP (20) corresponds to one surface. Therefore, the two objectives in (20) has the same level for consideration. A natural idea is combining two objectives to one as follows

$$\begin{aligned}
& \min_{u_1, u_2, \gamma_1, \gamma_2, \eta_1, \eta_2, E} \sigma e^T E e + \lambda (\|u_1\|_1 + c_{11} \|K(A_1 E, A E) u_1 + \gamma_1 e\|_1 + c_{12} \|\eta_1\|_1) \\
& \quad + (1 - \lambda) (\|u_2\|_1 + c_{21} \|K(A_2 E, A E) u_2 + \gamma_2 e\|_1 + c_{22} \|\eta_2\|_1) \\
& \text{s.t.} \quad -(K(A_2 E, A E) u_1 + \gamma_1 e) \geq e - \eta_1, \quad \eta_1 \geq 0, \\
& \quad K(A_1 E, A E) u_2 + \gamma_2 e \geq e - \eta_2, \quad \eta_2 \geq 0, \\
& \quad E = \text{diag}(1 \text{ or } 0),
\end{aligned} \tag{21}$$

where $\lambda \in (0, 1)$ is a parameter to balance the two objectives of (20).

There are many ways to solve the MOMIPP [27,10,4,40,28]. In this paper, we hire alternate iterative greedy algorithm [23] to find a local optimal value. To be more specific, the solution of (21) is obtained by fixing the integer part E to get the other real part u_i and γ_i , $i = 1, 2$, which leads to solve two LPPs the same as L_1 -TWSVM in Section 3, and then fixing the real part to get the integer part which leads to exhaustively compute the objective for the possible E , alternately, until meet the terminate condition.

To make the greedy algorithm work, the feature selection matrix E should be initialized first. Here, we propose a strategy to initialize E instead of randomly initializing it in RFSVM [23] to make the algorithm more stable. Specifically, the value of each feature is computed after solving L_1 -TWSVM by

$$\begin{aligned}
\text{Value}_{\text{feature}}(i) = & \lambda (c_{11} \|K(A_1(i), A(i)) u_1 + \gamma_1 e\|_1 \\
& + c_{12} e^T (K(A_2(i), A(i)) u_1 + \gamma_1 e + e)_+) \\
& + (1 - \lambda) (c_{21} \|K(A_2(i), A(i)) u_2 + \gamma_2 e\|_1 \\
& + c_{22} e^T (e - K(A_1(i), A(i)) u_2 - \gamma_2 e)_+),
\end{aligned} \tag{22}$$

where $(\cdot)_+$ replaces the negative elements with zeros, $A_1(i)$ means the i th column of A_1 , so does the $A_2(i)$ and $A(i)$. The score of the i th feature that reflects the importance among all the features is computed by

$$\text{Score}_{\text{feature}}(i) = \frac{\text{Value}_{\text{feature}}(i)}{\sum_j \text{Value}_{\text{feature}}(j)}. \tag{23}$$

Because our FTSVM aims at minimizing (21) by the selected features denoted by E , a feature has a larger value by (22) (or its percentage of all features by (23)) would actually have additional effect on the minimum of (21). Note that for the selected features, $\sigma e^T E e$ in (21) is a constant, and then this part is ignored in (22).

Our greedy algorithm starts from an initial $E^{(0)}$ generated by (23) (if $\text{Score}_{\text{feature}}(i)$ is less than a certain threshold, typically $1/n$, then $E_{ii}^{(0)} = 0$, otherwise $E_{ii}^{(0)} = 1$), then one can easily solve (21) for the fixed $E^{(0)}$ by solving (20). When obtained $(u_1^{(0)}, \gamma_1^{(0)}, u_2^{(0)}, \gamma_2^{(0)})$, the matrix $E^{(1)}$ is updated by replacing every diagonal element of $E^{(0)}$ from 1 to 0 (or 0 to 1) if the objective of (21) decreases more than the tolerance. In our greedy algorithm, we set E changed n times from the first diagonal element to the end if the objective of (21) decreased less than the tolerance, else return to change the first element to the end, until none of the diagonal element in E can be changed to make the objective of (21) decrease more than the tolerance. After updated E , $(u_1, \gamma_1, u_2, \gamma_2)$ can be updated again. The algorithm will be terminated if the objective of (21) decreased less than the tolerance. Using the sign $(\cdot)_+$, (21) can be simplified when a new $(u_1, \gamma_1, u_2, \gamma_2)$ is obtained as follows

$$\begin{aligned}
f(E) = & \sigma e^T E e + \lambda (c_{11} \|K(A_1 E, A E) u_1 + \gamma_1 e\|_1 \\
& + c_{12} \|(e + K(A_2 E, A E) u_1 + \gamma_1 e)_+\|_1) + (1 - \lambda) \\
& \times (c_{21} \|K(A_2 E, A E) u_2 + \gamma_2 e\|_1 \\
& + c_{22} \|(e - K(A_1 E, A E) u_2 - \gamma_2 e)_+\|_1).
\end{aligned} \tag{24}$$

We summarize the procedures of the greedy approach in Algorithm 1 to give our feature selection method.

Algorithm 1. Feature selection method for L_1 -TWSVM (FTSVM).

Input: The training set $A \in R^{m \times n}$. The appropriate parameters $c_{11}, c_{12}, c_{21}, c_{22}, \sigma$ (and a kernel function for nonlinear case). Pick a fixed integer k , typically very large, for the number of sweeps through E , and a stopping tolerance tol , typically $1e - 6$.

Output: $u_1, \gamma_1, u_2, \gamma_2, E$.

- 1: Set $E = I$, solve the LPPs (20) with the fixed E ;
- 2: Compute each feature score by (23), for $i = 1, \dots, n$, if $\text{Score}_{\text{feature}}(i) < 1/n$, then set $E_{ii} = 0$.
- 3: Solve the LPPs in (20) with the fixed E , and denote its solution by $(u_1, \gamma_1, u_2, \gamma_2)$.
- 4: **repeat**
- 5: For $l = 1, 2, \dots, kn$ and $j = 1 + (l - 1) \bmod n$:
 - (a) Replace E_{jj} by 0 if it is 1 and by 1 if it is 0.
 - (b) Compute the value (24) before and after changing E .
 - (c) Keep the new E_{jj} only if the objective function (24) decrease more than tol . Else undo the change of E_{jj} , and goto (a) if $j < n$.
 - (d) Goto 5 if the total decrease in (24) is less than or equal to tol in the last n steps.
- 6: Solve the LPPs in (20) with the new E , and compute the objective function (21) by the new solution $(u_1, \gamma_1, u_2, \gamma_2)$.
- 7: **until** the objective function decrease of (21) is less than tol if the solution $(u_1, \gamma_1, u_2, \gamma_2)$ is changed

The time complexity of our approach to solve the MOMIPP (20) includes two parts, one is repeatedly computing (24) (step 5(b) in Algorithm 1) and the other is solving many (20) (step 6 in Algorithm 1). For the first part, it is easy to compute with the fixed $(u_1, \gamma_1, u_2, \gamma_2)$, $f(E)$ is computed no more than k times. For the other part, suppose it needs to solve t LPPs to converge a solution, the time complexity of this part is at most $O(t2^{mn+1})$ by using the well known simplex method [5] to solve the LPPs. Therefore, the time complexity of our approach is at most $O(t2^{mn+1})$. The number of the LPPs t would be decreased if the parameter tol is set larger, and in our experiments, we set $\text{tol} = 1e - 6$ and get a good result with the tolerable time consuming, anyone can increase tol to get a faster learning speed or decrease tol to get the more favorable features.

4.3. RTSVM

In the following, we discuss the feature ranking (RTSVM) method based on L_1 -TWSVM. As discussed in Section 4.1, FTSVM searches for the optimal subspace spanned by the selected features that hyperplane is the closest to one class and the farthest from the other class in that subspace. One feature is selected and is important if it helps the classifier to find the optimal subspace, and the importance can be computed by (23). Owing to the contribution of the procedures of FTSVM, the selected features can be straightly ranked by (23), which is summarized in Algorithm 2.

Algorithm 2. Feature ranking method for L_1 -TWSVM (RTSVM).

Input: The training set $A \in R^{m \times n}$. The appropriate parameters $c_{11}, c_{12}, c_{21}, c_{22}, \sigma$ (and a kernel function for nonlinear case). The feature selection matrix E .

Output: The scores of the features.

- 1: Solve each problem in (20) with the fixed E ;
- 2: For $l = 1, 2, \dots, n$, Compute each feature score by (23).

5. Experimental results

In this section, we analyze the performance of our RTSVM and FTSVM (The matlab is uploaded upon <http://www.optimal-group.org/Resource/FSTWSVM.html>). We compare our implementation with state of the art L_1 -TWSVM and L_2 -TWSVM (The matlab code is uploaded upon <http://www.optimal-group.org/Resource.html>) on the ability of feature selection and generalization on several UCI datasets [1] (listed in Table 2). All these methods are implemented by Matlab [25] on a PC with an Intel P4 processor (3.4 GHz) with 4 GB RAM. For brevity's sake, we select the parameters $c_{11} = c_{21} \in \{2^i | i = -8, -7, \dots, 7\}$ and $c_{12} = c_{22} \in \{2^i | i = -8, -7, \dots, 7\}$ for the TWSVMs and our methods, the kernel parameter $\mu \in \{2^i | i = -10, -9, \dots, 5\}$ of the Gaussian kernel $K(x_i, x_j) = \exp\{-\mu \|x_i - x_j\|^2\}$ [36] for the nonlinear case for all these methods, and the parameter λ is selected from $\{0.1, 0.2, \dots, 0.9\}$ for our linear FTSVM, and for our nonlinear FTSVM we set $\lambda = 0.5$. If no particular claim, the 10-fold cross-validation methodology [17] is applied on the whole training set to estimate the generalized accuracy. The parameters of these methods with the highest accuracies are selected by grid search listed in Table 3.

In the first test, we focus on our RTSVM. First, for each dataset, the feature scores are computed by RTSVM and the features are discarded one by one according to the scores. Fig. 1 records the accuracies after discarding the features one by one for linear and nonlinear cases. Once the accuracy is lower than 50%, the experiment is terminated. From Fig. 1, it is easy to see that the accuracy almost cannot be higher than the original data when several features are ignored, i.e., RTSVM cannot help L_1 -TWSVM to select features so that perform well on these datasets. Second, the feature

Table 2
Details of the UCI datasets.

Data	Sample (No.)	Feature (No.)
Australian (Aus)	690	14
Echocardiogram (Ech)	131	10
Heartstatlog (Hea)	270	13
Hepatitis (Hep)	155	19
Ionosphere (Ion)	351	33
Sonar (Son)	208	60
Spect (Spe)	267	44
WPBC (Wpb)	198	34

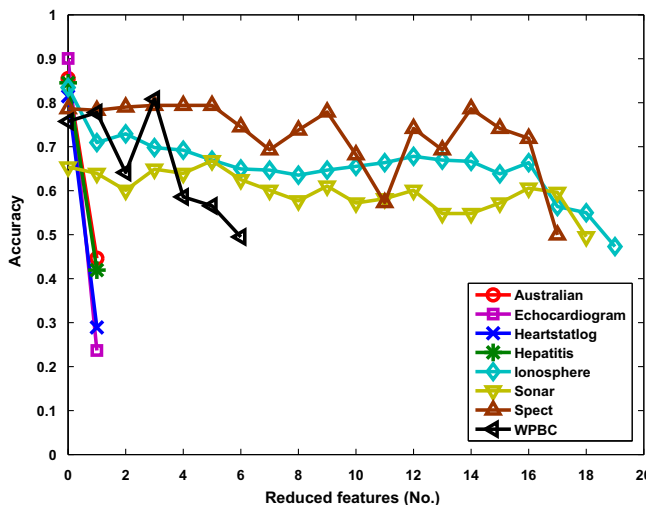
Table 3

The parameters of linear L_1 -TWSVM (M1: $c_{11} = c_{21}$, $c_{12} = c_{22}$), L_2 -TWSVM (M2: $c_{11} = c_{21}$, $c_{12} = c_{22}$), FTSVM (M3: $c_{11} = c_{21}$, $c_{12} = c_{22}$, λ), and nonlinear L_1 -TWSVM (N1: $c_{11} = c_{21}$, $c_{12} = c_{22}$, μ), L_2 -TWSVM (N2: $c_{11} = c_{21}$, $c_{12} = c_{22}$, μ), and FTSVM (N3: $c_{11} = c_{21}$, $c_{12} = c_{22}$, μ) with $\sigma = 0$ on benchmark datasets by grid search.

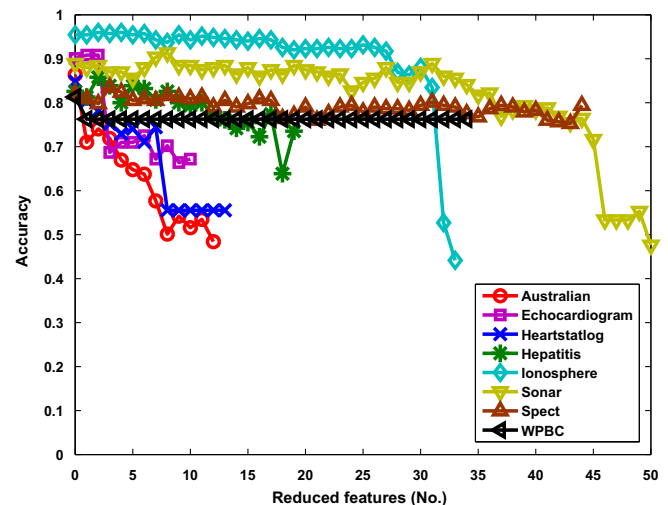
Data	M1	M2	M3	N1	N2	N3
Aus	$2^{-2}, 2^{-5}$	$2, 2^{-8}$	$2^{-4}, 2^6, 0.1$	$2^{-1}, 2^2, 2^{-3}$	$2^{-7}, 2^0, 2^{-9}$	$2^3, 2^5, 2^{-8}$
Ech	$2^{-4}, 2^{-8}$	$2^{-8}, 2^0$	$2^{-2}, 2^4, 0.1$	$2^2, 2^3, 2^{-1}$	$2^{-8}, 2^{-7}, 2^{-8}$	$2^3, 2^4, 2^{-3}$
Hea	$2^{-8}, 2^0$	$2, 2^{-2}$	$2^{-7}, 2^{-8}, 0.4$	$2^3, 2^3, 2^{-10}$	$2^{-8}, 2^{-3}, 2^{-8}$	$2^2, 2^2, 2^{-9}$
Hep	$2^{-4}, 2^{-5}$	$2^0, 2$	$2^{-7}, 2^{-6}, 0.3$	$2^0, 2^2, 2^{-4}$	$2^0, 2^{-8}, 2^{-8}$	$2^0, 2^3, 2^0$
Ion	$2^{-4}, 2^{-5}$	$2^{-5}, 2^{-4}$	$2^{-4}, 2^{-1}, 0.5$	$2^0, 2^3, 2^{-4}$	$2^{-7}, 2^{-6}, 2^{-2}$	$2^0, 2^3, 2^{-2}$
Son	$2^{-2}, 2^3$	$2^3, 2^{-1}$	$2^{-8}, 2^{-6}, 0.3$	$2^6, 2^3, 2^{-3}$	$2^{-7}, 2^{-8}, 2^{-3}$	$2, 2^5, 2^{-2}$
Spe	$2^{-7}, 2^{-6}$	$2^{-4}, 2^0$	$2^{-3}, 2^{-5}, 0.1$	$2, 2^3, 2^{-2}$	$2^{-2}, 2^{-6}, 2^{-6}$	$2, 2^4, 2^0$
Wpb	$2^7, 2^6$	$2^{-2}, 2^{-7}$	$2^{-1}, 2^{-3}, 0.3$	$2^7, 2^7, 2^{-6}$	$2^{-3}, 2^{-7}, 2^{-4}$	$2^5, 2^6, 2^{-5}$

set of each dataset is divided into three parts with almost the same number of features, where the top, middle and bottom parts have the top, middle and bottom scores by RTSVM, respectively. We test the performance of L_1 -TWSVM on these subsets of features and record the results in Fig. 2. From Fig. 2, it is obvious that the top-score subset almost always performs better than the bottom-score subset, which indicates our RTSVM can do nice feature ranking.

In the second test, we focus on our FTSVM on the generalization ability by setting $\sigma = 0$, i.e., we pay more attention on getting an efficient classifier rather than reducing the number of selected features. The 10-fold cross-validation accuracy and the number of feature selected of our FTSVM compared with L_1 -TWSVM and L_2 -TWSVM are implemented in this test. For the linear case, the feature selection by linear L_1 -TWSVM is also implemented, for it can select features when the corresponding elements of w_1 and w_2 both equal to zero in (1), and the feature would not be selected if either $w_1 > 0$ or $w_2 > 0$. We implement each dataset ten times and record the average number of feature selected, the training time and the average accuracy with the standard deviation in Table 4 for linear and nonlinear cases, the highest accuracies and the smallest number of features are bold. It is obvious that our FTSVM always can reduce more features without any loss generalization (having the highest accuracy on 6/8 datasets), though L_1 -TWSVM can select less features on several datasets (1/8) in Table 4, it reduces a little features on the other datasets (7/8), and it cannot reduce any feature for the nonlinear case. The linear



(a) Linear case



(b) Nonlinear case

Fig. 1. The performance of L_1 -TWSVM on benchmark datasets by reducing features one by one according to the score of RTSVM.

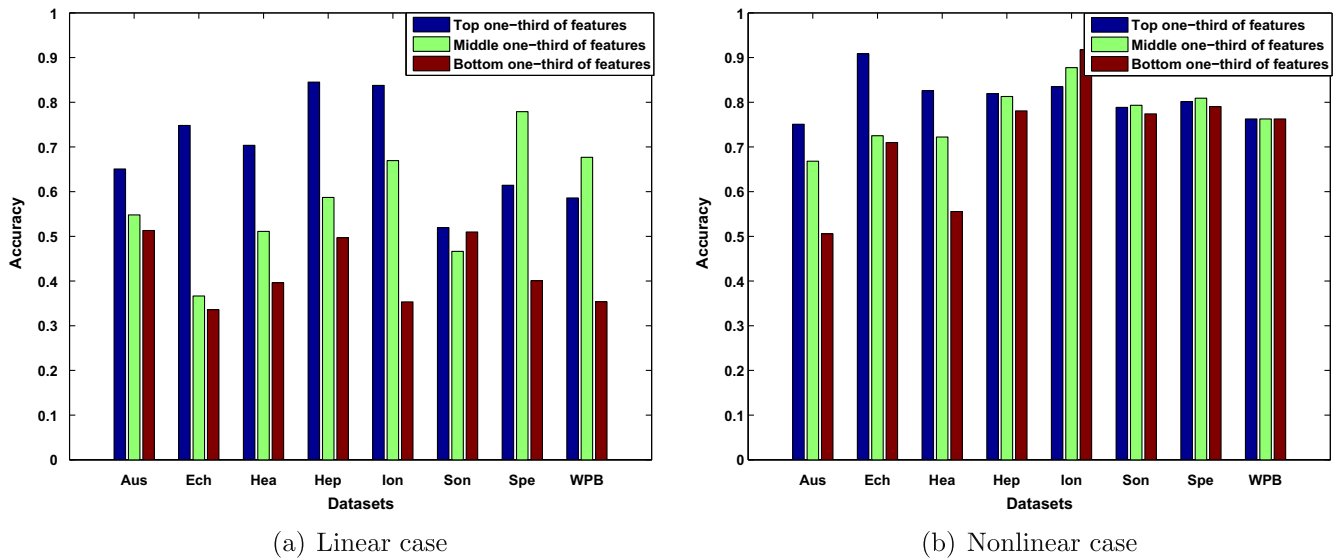


Fig. 2. The performance of L_1 -TWSVM on benchmark datasets on the subsets of features by the score from RTSVM (for each dataset, left bar stands for using top one-third features, middle bar stands for middle one-third features and right bar stands for bottom one-third features).

Table 4
Results (accuracy $\pm$ std, feature/time) of linear L_1 -TWSVM (M1), L_2 -TWSVM (M2), FTSVM (M3), and nonlinear L_1 -TWSVM (N1), L_2 -TWSVM (N2), and FTSVM (N3) with $\sigma = 0$ on benchmark datasets.

Data	M1	M2	M3	N1	N2	N3
Aus	85.38 $\pm$ 0.29 11.7/0.0209	86.61 $\pm$ 0.20 14/0.9937	80.30 $\pm$ 5.10 1/2.262	85.72 $\pm$ 0.86 14/3.6715	87.03 $\pm$ 0.34 14/0.2973	86.38 $\pm$ 1.73 8/244.1
Ech	88.17 $\pm$ 3.51 2.4 /0.0050	91.30 $\pm$ 0.74 10/0.0130	85.50 $\pm$ 0 4/0.3492	88.55 $\pm$ 1.44 10/0.0299	90.92 $\pm$ 0.76 10/0.0165	91.07 $\pm$ 1.08 7/4.424
Hea	79.78 $\pm$ 3.25 11.7/0.0320	83.52 $\pm$ 0.77 13/0.1403	79.00 $\pm$ 0.86 11 /2.205	84.29 $\pm$ 0.53 13/0.2081	83.70 $\pm$ 0.49 13/0.0387	84.63 $\pm$ 0.47 12/28.50
Hep	83.35 $\pm$ 1.89 18.9/0.0153	83.87 $\pm$ 0.68 19/0.0859	81.35 $\pm$ 1.91 12/0.5366	83.81 $\pm$ 0.83 19/0.0419	84.06 $\pm$ 0.81 19/0.0901	86.45 $\pm$ 0.86 10 /7.256
Ion	83.68 $\pm$ 0.27 32.8/0.1985	91.99 $\pm$ 0.68 33/0.0728	84.13 $\pm$ 2.76 29/4.577	95.64 $\pm$ 0.45 33/0.5460	95.95 $\pm$ 0.48 33/0.1640	96.38 $\pm$ 0.40 17 /66.99
Son	63.27 $\pm$ 2.28 59.8/0.0399	75.67 $\pm$ 1.38 60/0.0747	68.03 $\pm$ 2.21 11 /1.713	88.12 $\pm$ 1.43 60/0.1056	89.28 $\pm$ 1.36 60/0.0724	89.33 $\pm$ 0.79 23/45.78
Spe	78.99 $\pm$ 0.33 44/0.0286	80.19 $\pm$ 0.80 44/0.0797	79.40 $\pm$ 0 40/2.031	81.87 $\pm$ 1.06 44/0.3924	83.48 $\pm$ 0.57 44/0.1202	81.31 $\pm$ 1.22 38 /28.82
Wpb	78.89 $\pm$ 2.54 34/0.0261	81.26 $\pm$ 1.94 34/0.0107	72.27 $\pm$ 4.28 33/1.545	79.95 $\pm$ 0.68 34/0.1140	81.21 $\pm$ 1.40 34/0.0939	81.41 $\pm$ 0.62 22 /12.13
W/L	0/8	0/8	0/8	0/8	2/8	6/8

Table 5
Results (accuracy $\pm$ std, feature/time) of linear L_1 -TWSVM (M1), L_2 -TWSVM (M2), FTSVM (M3), and nonlinear L_1 -TWSVM (N1), L_2 -TWSVM (N2), and FTSVM (N3) with $\sigma = 0$ on two datasets with hundreds of features.

Data	M1	M2	M3	N1	N2	N3
Fbi	69.62 $\pm$ 3.84	74.32 $\pm$ 2.16	83.18 $\pm$ 2.87	77.95 $\pm$ 1.15	75.76 $\pm$ 0.62	83.33 $\pm$ 1.75
132 $\times$ 800	800/0.1125	800/0.0694	479/32.69	800/0.0266	800/0.0980	432 /1487
Arcene	60.90 $\pm$ 4.12	77.40 $\pm$ 2.67	80.90 $\pm$ 3.31	78.70 $\pm$ 1.83	78.00 $\pm$ 1.41	90.40 $\pm$ 1.07
100 $\times$ 1000	1000/0.1052	1000/0.1927	67/19.89	1000/0.0147	1000/0.0414	412/1553

L_2 -TWSVM gets higher accuracy compared with the linear classifier L_1 -TWSVM and FTSVM, while nonlinear FTSVM gets the highest accuracy on most datasets (6/8) compared with other linear and nonlinear classifiers. In a word, our FTSVM can actually improve the performance of TWSVM on the generalization ability with less features, especially for the nonlinear case. Moreover, two datasets [26,1] with hundreds of features are also tested by these methods

and the results are reported in Table 5. Note that each feature weight of linear L_1 -TWSVM on these two datasets belongs to $(1e - 14, 1e - 15)$, so none of features can be ignored. Obviously, our linear and nonlinear FTSVMs perform better than others with lesser features.

In the third test, we focus on the feature selection ability of our FTSVM by adjusting the parameter σ . We select σ from

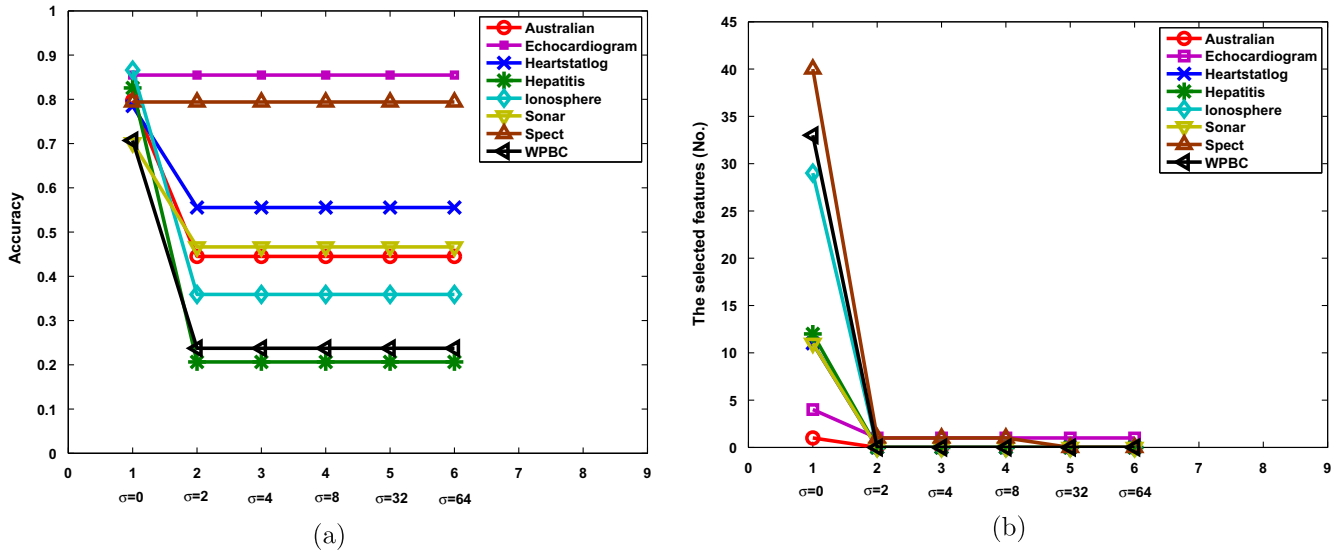


Fig. 3. The performance of linear FTSVM on benchmark datasets with different σ .

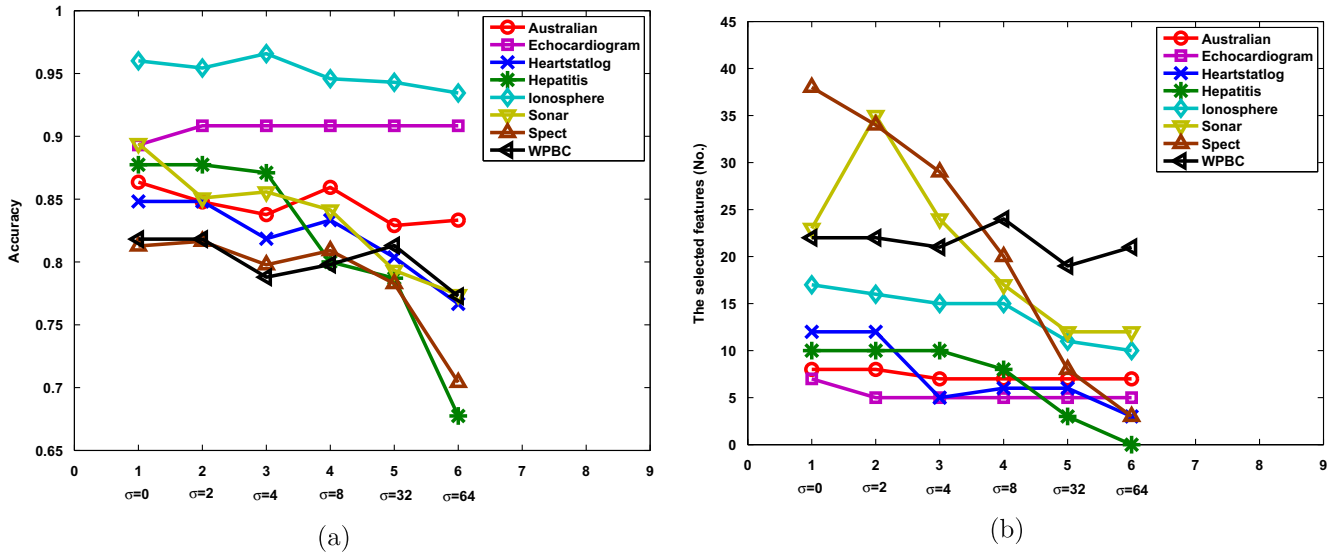


Fig. 4. The performance of nonlinear FTSVM on benchmark datasets with different σ .

$\{0, 2, 4, 8, 32, 64\}$ and test our FTSVM on these UCI datasets for linear and nonlinear cases. For each value of σ , we record the accuracy and the number of feature selected in Figs. 3 and 4 for linear and nonlinear cases, respectively. It is obvious that lesser features are selected with higher σ , and meanwhile the accuracy may decrease more. From Fig. 3, it is known that any $\sigma > 0$ would actually decrease the performance of the FTSVM. However, for nonlinear case in Fig. 4, the parameter σ plays an important role in the method, in fact, one can increase σ to obtain less features if necessary. As a conclusion, the parameter σ can control the number of the selected features for nonlinear case, and the highest accuracy often is obtained when $\sigma = 0$, i.e., when the method aims at getting the highest generalization ability and do not care about how many features are selected.

6. Conclusions

In this paper, we have proposed a novel feature selection method (FTSVM) based on TWSVM with L_1 norm regularization.

Different from L_1 -SVM with only a single feature weight, the corresponding L_1 -TWSVM has two feature weights, resulting some difficulty for feature selection. Thus, a feature selection matrix is introduced, aiming at realizing feature selection. This approach leads to solve a multi-objective mixed integer programming problem by an alternate iterative greedy algorithm. As an adjunct to the feature selection, a feature ranking method (RTSVM) is also proposed. It is interesting to point out that different from many feature selection and ranking methods, both our FTSVM and RTSVM have the linear formations as well as the nonlinear ones. The preliminary experiments have demonstrated the efficiency of our FTSVM and RTSVM, especially for the nonlinear case. For practical convenience, we upload our FTSVM Matlab code upon <http://www.optimal-group.org/Resource/FSTWSVM.html>.

For the further work, it is interesting to extend our FTSVM and RTSVM to the relevant nonparallel hyperplanes classifiers without any essential difficulties, e.g., GEPSVM [24], RTWSVM [31], NHSVM [29], STPMSVM [42], LSTPMSVM [34] and PPSVC [41]. Another practical topic is to find more efficient method to solve the

MOMIPPs (18) and (20) since the current method is time consuming in some sense.

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